EEN1095 Project Progress and Change Log

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Project Title	Yield-Aware Inverse Design of High-Speed Interconnects: A Physics-Constrained Generative Approach	**Supervisor**	Dr. Marissa Condon
Implementation Period	Week 9 - Week 10	**Version Date**	16/07/2026

Guidance: Maintain this as one **cumulative** document throughout EEN1095. Add a new entry every two weeks and do not remove earlier records. Progress should be recorded in a timely and concise manner. Where no change has occurred, this should be stated explicitly.

1. Project Overview

Original approved project title	Yield-Aware Inverse Design of High-Speed Interconnects: A Physics-Constrained Generative Approach
Original project objectives	Build an yield aware inverse model for the high-speed interconnects to predict the geometries from the target s-parameters
Original proposed methodology	Tandem network with forward model as a validator and inverse model predicting geometries with Jacobian regularization for yield optimization
Planned outputs / deliverables	Yield optimized inverse model predicting the geometries from the input s-parameter and global features.

2. Summary of Key Milestones

The table below is intended as a high-level summary. Use Section 3 for the detailed bi-weekly updates.

Period	Planned activity	Progress made	Outcome / status
Weeks 1-2	Complete the Forward model, start working on Inverse design	Forward model completed, inverse design fundamental design steps and execution started	Data Pipeline – completed Forward Model – Completed (partially) Inverse design – In-Progress

Weeks 3-4	Design the baseline architecture for the inverse design model and try implementing yield optimization		completed
Weeks 5-6	Improvise the inverse model, EDA to study frequency regions, openems implementation.	Completed the baseline design of the inverse model.	Completed
Weeks 7-8	Latent Space TTO, openEMS tutorial simulation	TTO completed, openEMS in progress	Completed
Weeks 9-10	openEMS simulation and validation of geometry generated by the model.	openEMS simulation and validation complete	Completed
Weeks 11-12			[Completed / In progress / Revised]
Weeks 13-14			[Completed / In progress / Revised]

3. Detailed Progress Entries

Entry 1: Weeks 1-2

Main objectives for this period	Complete Forward Model, start inverse design
Work completed	Forward model trained and completed with average MAE of 2.12dB on a complex TUHH dataset
Key outputs produced	Forward model with MAE of 2.12dB tracking the curve.
Problems encountered	Forward Model tracks the resonance dip (struggles a bit on 55GHz and above range)
Changes made to plan or methodology	The Data pipeline logic was changed to extract the 4x4 matrix of all differential pairs to augment the dataset and increase the datapoints available. Changed from rational forward net to 1D CNN – Direct sequence ResNet
Reasons for any changes	Data-pipeline change was to increase the no of datapoints. Forward model change was CNN performed better than Rational Model
Next steps	Working on the inverse model

Entry 2: Weeks 3-4

Main objectives for this period	Design and implement the inverse CVAE architecture in the tandem network with frozen weights from the forward model
Work completed	Implemented tandem network successfully and connected the baseline cvae design to link with the frozen weight from the forward model.
Key outputs produced	The baseline model predicts the geometries from the given s-parameter profile and the forward model validates the generated design to keep the geometries in the desired scope obeying physical constraints.
Problems encountered	Strict penalty functions making the curve to focus on the low frequency range and ignoring the resonance dips in the high frequency range.
Changes made to plan or methodology	No changes in the plan – the baseline CVAE architecture was implemented and the tandem network was connected.
Reasons for any changes	N/A
Next steps	Improvise the inverse model – implement yield optimization and adding the active learning loop to the pipeline.

Entry 3: Weeks 5-6

Main objectives for this period	Improvise the inverse model and perform EDA on the frequency range to study the importance of the regions in the TUHH SI/PI database
Work completed	EDA on the frequency range and initial baseline inverse model design with test time optimization technique to check if accuracy is improved.
Key outputs produced	The EDA says that for sd11 the primary focus of the design is on low frequency and, it is exact opposite for sdd21 where it is on the high frequency range. Test Time optimization improves the accuracy of the model when compared to the actual prediction.
Problems encountered	Resonance dips were missed because of the weights provided on text book logic in the loss function, performed EDA on the frequency range to provide masked regions and allot weights in the loss function accordingly.
Changes made to plan or methodology	Implemented the baseline with normal cVAE architecture, changed the weights in the loss function based on the EDA performed on the freq range studying the importance of the regions.
Reasons for any changes	The model was tracking noise region in the parameter curve and missing the deep resonance dips so I tried to ddo an EDA to study the importance of the frequency regions

Next steps	Improvise the inverse model to be more accurate, and implement openEMS simulation to validate the output generated by the inverse model.

Entry 4: Weeks 7-8

Main objectives for this period	Test time optimization to the inverse model
Work completed	Completed the implementation of test time optimization for the inverse model (now working on openEMS and also latent TTO)
Key outputs produced	The test time optimization improved the accuracy of the inverse model compared to the traditional cVAE arch.
Problems encountered	The insertion loss curve had a mismatch of around 2dB while trying to fit the curve (this is because of surrogate exploitation)
Changes made to plan or methodology	Improvement to be made to the TTO by using Latent Space TTO and cumulative TTO to the model
Reasons for any changes	The initial implementation of the TTO was focused on the geometry but it had surrogate exploitation as it focused on geometries that are physically invalid. This to be improved using Latent Space TTO (focusing on latent space) and also with passivity in it for physically valid results.
Next steps	Work on openEMS simulation and validate the results generated by the model. Implement Cumulative TTO (Latent Space and passivity)

Entry 5: Weeks 9-10

Main objectives for this period	OpenEMS simulation and Validation
Work completed	Completed openEMS simulation and validated the results of ground truth and the model generated geometry.
Key outputs produced	The model generated geometry matched with the s-parameter profile almost close to the ground truth data for the same s-parameter profile.
Problems encountered	The simulation was crashed and exploded due to the use of lumped ports (PML_8) but it was rectified with using co-axial ports in the openEMS.
Changes made to plan or methodology	N/A
Reasons for any changes	N/A

Next steps	Improvise Latent & Cummulative TTO, and implement yield optimization to the model.

Entry 6: Weeks 11-12

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]
Next steps	[Insert concise notes here]

Entry 7: Weeks 13-14

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]
Next steps	[Insert concise notes here]